

DIFFERENTIAL EQUATION

Differential Equation:

The equation involving derivative (differential) is called differential equation. For example

$$\frac{dy}{dx} + y \cos x = \sin x$$

Order of the Differential Equation:

The order of the differential equation is the order of the highest derivative that occurs in the equation.

Degree of the differential Equation:

The degree of a differential equation is the greatest exponent of the highest order derivative that appears in the equation.

Following table shows the order and degree of some differential equation.

Differential Equation	Order	Degree
$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx}\right) = 0$	2	1
$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = \frac{d^2y}{dx^2}$	2	2
$\frac{dy}{dx} = -ky$	1	1

Solution of the Differential Equation

Equation free from derivative is called the solution (general solution) of that differential equation. A solution derived from a general solution is called particular solution.

Ordinary Differential Equation:

A differential equation, in which ordinary derivatives of the dependent variable with respect to a single independent variable occur, is called an ordinary differential equation. For example.

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$$

Partial Differential Equation:

A differential equation involving partial derivatives of the dependent variable with respect to more than one independent variable is called a partial differential equation.

For example $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

Linear Differential Equation:

A linear differential equation is any differential equation that can be written in the following form.

$$a_n(t)y^{(n)}(t) + a_{n-1}(t)y^{(n-1)}(t) + \cdots + a_1(t)y'(t) + a_0(t)y(t) = g(t)$$

The important thing to note about linear differential equations is that there are no products of the function $y(t)$, and its derivatives and neither the function or its derivatives occur to any power other than the first power. The coefficients $a_0(t), \dots, a_n(t)$ and $g(t)$ can be zero or non-zero functions, constant or non-constant functions, linear or non-linear functions. Only the function, $y(t)$, and its derivatives are used in determining if a differential equation is linear.

If a differential equation cannot be written in the above form then it is called a non-linear differential equation.

Some Types of First Order Differential Equations:

Following table shows the information of some standard first order differential equations.

Separable	$F(x) G(y)dx + P(x)Q(y)dy = 0$
Homogenous	$M(x, y)dx + N(x, y) dy = 0$ where $M(x, y)$ and $N(x, y)$ are homogeneous functions of same degree.
Exact	$M(x, y) dx + N(x, y) dy = 0$ Where $M_y = N_x$
Linear	$\frac{dy}{dx} + P(x)y = Q(x)$
Bernoulli	$\frac{dy}{dx} + P(x)y = Q(x)y^n, n > 1.$
Clairauts	$y = xP + f(P)$ where $P = \frac{dy}{dx}$
Ricatti	$y' + P(x)y + Q(x)y^2 = R(x)$

Initial Condition:

The problem of finding the solution of a differential equation such that all the associated constraints relate to one value of the independent variable is called a one-point boundary value problem (initial value problem). The associated called supplementary condition is boundary initial value condition.

Boundary Conditions:

The problems of finding the solution of a differential equation such that all the associated constraints relate to two different values of the independent variable is called a two-point boundary value problems. (boundary value problem). The associated supplementary boundary conditions are called two point boundary conditions.

Some Important Differential Equations:

Following table shows the information of standard differential equations of every day life.

Laplace Equation	$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$
Heat Equation	$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} - k \frac{\partial u}{\partial t} = 0$
Wave Equation	$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} - \frac{1}{C^2} \frac{\partial^2 u}{\partial t^2} = 0$
Legendre Equation	$(1 - x^2) \frac{dy}{dx} - 2x \frac{dy}{dx} + l(l+1)y = 0$
Bessel Equation	$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$
Hermite Equation	$\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2ny = 0$
Laguerre Equation	$x \frac{d^2 y}{dx^2} + (1 - x) \frac{dy}{dx} + ny = 0$
Chebyshev Equation	$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + n^2 y = 0$
Gegenbauer Equation	$(1 - x^2) \frac{d^2 y}{dx^2} - (2m+1)x \frac{dy}{dx} + n(n+2m)y = 0$
Hyper Geometric Equation	$x(1-x) \frac{d^2 y}{dx^2} + \{r - (\alpha + \beta + 1)x\} \frac{dy}{dx} - \alpha\beta y = 0$
Whittaker's Confluent Hypergeometric Equation	$\frac{d^2 y}{dx^2} + \left\{ -\frac{1}{4} + \frac{k}{x} + \frac{\frac{1}{4} - n^2}{x^2} \right\} y = 0$
Harmonic Oscillator Equation	$\frac{d^2 y}{dx^2} + w^2 y = 0$
Airy's Equation	$\frac{d^2 y}{dx^2} - xy = 0$
Euler Equation	$x^2 \frac{d^2 y}{dx^2} + ax \frac{dy}{dx} + \beta y = 0, x > 0$
Gompertz Equation	$\frac{dy}{dt} = \gamma y \ln\left(\frac{k}{y}\right)$
Verhulst Equation	$\frac{dy}{dt} = (r - ay)y$

Multiple Choice Questions:

1. Equation involving derivative is called:
(a) differential equation (b) integral equation
(c) derive equation (d) all of these
2. A differential equation in which ordinary derivatives of the dependent variable with respect to a single independent variable occur, is called:
(a) ordinary differential equation (b) partial differential equation
(c) homogenous differential equation (d) none
3. If in a differential equation, the dependent variable and its derivative are all of degree one and no product term of dependent variable and its derivative occur, then differential equation is:
(a) partial differential equation (b) linear differential equation
(c) homogeneous differential equation (d) none
4. The power of the highest order derivative that appear in the differential equation is called:
(a) order of differential equation (b) degree of differential equation
(c) power of differential equation (d) length of differential equation
5. The degree of differential equation $\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx}\right)^2 = 0$ is:
(a) 0 (b) 1
(c) 2 (d) 3
6. The order of differential equation $\left[\frac{dy}{dx}\right]^{3/2} = \frac{d^2y}{dx^2}$ is:
(a) 0 (b) 1
(c) 2 (d) 3
7. The degree of differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = \frac{d^2y}{dx^2}$ is:
(a) 0 (b) 1
(c) 2 (d) 3
8. The general solution of differential equation contains arbitrary constant equal to the:
(a) order of differential equation (b) highest order derivative
(c) degree of differential equation (d) all of these
9. The solution derive from the general solution is called:
(a) initial solution (b) particular solution
(c) boundary value solution (d) none
10. If supplementary conditions of a differential equation relate to one value of the independent variable then these conditions are:
(a) boundary conditions (b) space variable conditions
(c) initial conditions (d) mixed conditions
11. The boundary conditions of differential equations relate to how many value of the independent variable.
(a) 1 (b) 2

- (c) 3 (d) 4
12. Which of the following is a differential equation of all circle of radius a .
- (a) $a^2 \frac{d^2 u}{dx^2} = \frac{d^2 u}{dy^2}$ (b) $2a \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0$
- (c) $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = a \frac{d^2 y}{dx^2}$ (d) none
13. The equation $xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$ represents:
- (a) circles that passes through origin (b) ellipse in standard form
(c) hyperbolas in standard form (d) parabolas in standard form
14. Which of the following is separable equation.
- (a) $(xy + 2x + y + 2) dx + (x^2 + 2x) dy = 0$
(b) $y \sqrt{1 + x^2} dx + x \sqrt{1 + y^2} dy = 0$
(c) $xe^{x+y} dx = y dy$ (d) all of these
15. Which of the following is not separable equation.
- (a) $\frac{dy}{dx} = y^2 \sin x$ (b) $\sec y dx + \sec x dy = 0$
(c) $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$ (d) $(x^2 + 3y^2) dx - 2xy dy = 0$
16. Which of the following is a solution of $\frac{dx}{x} = \tan y dy$.
- (a) $x \sec y = c$ (b) $\sin y \cos y = x + a$
(c) $x \cos y = a$ (d) $\sec x + \sec y = b$
17. If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $f(tx, ty) = t^n f(x, y)$ then f is homogeneous function of degree
- (a) t (b) $n - 1$
(c) n (d) $n + 1$
18. The function $f(x, y) = \sqrt{xy}$ is homogeneous of degree
- (a) 0 (b) 1
(c) 2 (d) $\frac{1}{2}$
19. The function $f(x, y) = \sin\left(\frac{x}{y}\right)$ is homogeneous of degree
- (a) 0 (b) 1
(c) 2 (d) 3
20. Which of the following function is homogeneous of degree 4.
- (a) $\frac{x^4 y^4}{x + y}$ (b) $\frac{x^4 + y^4}{x + y}$
(c) $\frac{x^8 + y^8}{x^4 + y^4}$ (d) $\frac{x^6 y^6}{x^3 + y}$
21. Which of the following is homogeneous differential equation.

- (a) $(x - y) dx + (x + y) dy = 0$ (b) $y dy + x dx = \sqrt{x^2 + y^2} dx$
 (c) $\frac{dy}{dx} = \frac{x + y}{x}$ (d) all of these
22. If differential equation $(a_1x + b_1y + c_1) dx + (a_2x + b_2y + c_2) dy = 0$ is not homogeneous such that $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$, then which of the following is a suitable transformation to make it homogeneous.
 (a) $x = X - h, y = Y - k$ (b) $x = X + h, y = Y + h$
 (c) $x + X = h, y + Y = h$ (d) all of these
23. If differential equation $(a_1x + b_1y + c_1) dx + (a_2x + b_2y + c_2) dy = 0$ is not homogeneous such that $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then by replacing $Z = a_1x + b_1y$ its become
 (a) homogeneous (b) linear
 (c) separable (d) exact
24. The equation $M(x, y) dx + N(x, y) dy = 0$ is called exact differential equation if:
 (a) $M(x, y) = f_y$ (b) $M(x, y) = f_x$
 (c) $N(x, y) = f_x$ (d) $M(x, y) + N(x, y) = f_x - f_y$
25. The equation $M(x, y) dx + N(x, y) dy = 0$ is exact if:
 (a) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$ (b) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$
 (c) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$ (d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
26. Which of the following is exact differential equation.
 (a) $(1 + \ln xy) dx + \left(1 + \frac{x}{y}\right) dy = 0$ (b) $(y \cos x + 2xe^x) dx + (\sin x + x^2e^y - 1) dy = 0$
 (c) $(2xy - 3) dx + (x^2 + 4y) dy = 0$ (d) all of these
27. The differential equation $\frac{y dx + x dy}{1 - x^2y^2} + x dx = 0$ is:
 (a) linear (b) exact
 (c) separable (d) homogeneous
28. Which of the following is integrating factor for non-exact differential equation $M(x, y) dx + N(x, y) dy = 0$ if $\frac{My - Nx}{N} = P(x)$
 (a) $\int p(x) dx$ (b) $\int p(y) dx$
 (c) $\int p(y) dy$ (d) $\int p(x) dy$
29. If $M(x, y) dx + N(x, y) dy = 0$ is homogeneous such that $xM + yN \neq 0$, then to make it exact, which of the following is integrating factor.
 (a) $\int p(x, y) dx dy$ (b) $\frac{1}{xM + yN}$

- (c) $\frac{1}{xM + yN}$ (d) $\int p(y) dy$
30. Which of the following is equal to $xdy + ydx$.
(a) $d\left(\frac{x}{y}\right)$ (b) $d\left(\frac{y}{x}\right)$
(c) $d(xy)$ (d) $d(x + y)$
31. Which of the following is equal to $\frac{ydx - xdy}{xy}$.
(a) $d(x^2 + y^2)$ (b) $d\left(\ln \frac{x}{y}\right)$
(c) $d\left(\tan^{-1} \frac{x}{y}\right)$ (d) $d(xy)$
32. The general solution of $ydx + (x^2y - x) dy = 0$ is
(a) $xy^2 - 2y = cx$ (b) $y(xy - 2) = cx$
(c) $2y = x(y^2 - c)$ (d) all of these
33. The integrating factor for $dx + \left(\frac{x}{y} - \sin y\right) dy = 0$ is
(a) x (b) y
(c) e^x (d) $\ln(xy)$
34. Which of the following is not exact.
(a) $xdy - ydx = (x^2 + y^2) dx$ (b) $y - x \frac{dy}{dx} = x + y \frac{dy}{dx}$
(c) $\frac{dy}{dx} = e^{2x} + y - 1$ (d) all of these
35. If a first order ordinary differential equation of the form $\frac{dy}{dx} + p(x)y = Q(x)$ is called:
(a) linear (b) exact
(c) separable (d) bernoulli
36. Which of the following is linear differential equation.
(a) $\frac{dr}{d\theta} + r \tan \theta = \cos^2 \theta$ (b) $\frac{dy}{dx} + \frac{y}{x \ln x} = \frac{3x^2}{\ln x}$
(c) $x \frac{dy}{dx} + 2y = \sin x$ (d) all of these
37. The integrating factor for linear differential equation $\frac{dx}{dy} - \frac{x}{y} = 2y^3$ is:
(a) y (b) y^{-1}
(c) y^2 (d) $\ln y$
38. For a Bernoulli differential equation $\frac{dy}{dx} + p(x)y =$
(a) $Q(x)$ (b) $Q(x, y)$
(c) $Q(x)P(y)$ (d) $Q(x)y^n, n > 1$

39. Which of the following is Bernoulli differential equation.
- (a) $\frac{dy}{dx} + y = xy^3$ (b) $x \frac{dy}{dx} + y = y^2 \ln x$
- (c) $\frac{dy}{dx} = \frac{1}{e^y - x}$ (d) all of these
40. Each member of a one-parameter family of curves is a _____ of some differential equation.
- (a) order (b) degree
- (c) solution (d) root
41. Let $f(x, y, c) = 0$ and $F(x, y, k) = 0$ be two families of curve with parameters c and k . If each curve in either family is intersected orthogonally by every curve in other family, then each family is:
- (a) perpendicular to other (b) orthogonal trajectory of other
- (c) orthogonally perpendicular to other (d) all of these
42. Two curves are orthogonal if their tangents at the point of intersection are:
- (a) parallel (b) perpendicular
- (c) colinear (d) coplanar
43. Which of the following is orthogonal trajectories of the family of circles $x^2 + y^2 = c^2$
- (a) $x^2 - y^2 = k^2$ (b) $2x^2 + y = k$
- (c) $y = kx$ (d) none
44. A family of orthogonal trajectories is the same as given family is called:
- (a) point-orthogonal (b) single-orthogonal
- (c) self-orthogonal (d) symmetry
45. The orthogonal trajectory of $r = a \sin 2\theta$ is:
- (a) $r = a(1 - \sin \theta)$ (b) $r^2 = b \cos 2\theta$
- (c) $r \sin \theta = b$ (d) none
46. Which of the following is self orthogonal.
- (a) $xy = c$ (b) $y = e^{cx}$
- (c) $y^2 = 4cx + 4c^2$ (d) $r = a \sin n\theta$
47. The equation of the type $y = xp + f(p)$ where $p = \frac{dy}{dx}$ is:
- (a) Bernoulli equation (b) Linear equation
- (c) Clairaut's equation (d) Homogeneous equation
48. The locus of the points of intersection of neighbouring curves of a one-parameter family $f(x, y, c) = 0$ is called:
- (a) trajectory (b) family
- (c) envelop (d) system

49. The solution of a differential equation which is not derived from a general solution is called.
(a) particular solution (b) singular solution
(c) joint solution (d) extraneous solution
50. The c-discriminant equation may contain.
(a) general solution (b) particular solution
(c) singular solution (d) all of these
51. The p -discriminant represents the locus for each point of which $f(x, y, p) = 0$ has
(a) equal roots (b) distinct roots
(c) real roots (d) complex roots
52. The equation of the type $y' + P(x)y + Q(x)y^2 = R(x)$ is
(a) Bernoulli's equation (b) Ricatti equation
(c) Clairaut's equation (d) Linear equation
53. In the solution of higher order differential equation $y = y_c + y_p$ where y_c is complementary function, then y_p is:
(a) particular solution (b) particular integral
(c) point solution (d) none
54. If the characteristic equation has real and distinct roots m_1 and m_2 , then solution of differential equation is
(a) $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ (b) $y = c_1 e^{m_1 x} + e^{m_2 x}$
(c) $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ (d) $y = c_1 c_2 e^{m_1 x} e^{m_2 x}$
55. If roots m_1 and m_2 of characteristic equations are real and repeated then solution of differential equation is
(a) $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ (b) $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$
(c) $y = (c_1 + c_2 x) e^{m_1 x}$ (d) none
56. If the solution of differential equation is of the type
 $y = (c_1 + c_2 x) \sin bx + (c_3 + c_4 x) \cos bx$
then roots of characteristic equation are
(a) real and distinct (b) real and repeated
(c) complex and distinct (d) rational and distinct
57. Which of the following is a solution of $(D^2 + 4D + 3)y = 0$.
(a) $y = c_1 e^x + c_2 e^{2x}$ (b) $y = c_1 e^{-x} + c_2 e^{2x}$
(c) $y = c_1 e^{-x} + c_2 e^{-3x}$ (d) $y = c_1 e^{2x} + c_2 e^{4x}$
58. The solution of $(D^3 - D^2 + D - 1)y = 0$ is:
(a) $y = c_1 e^x + c_2 \sin x + c_3 \cos x$ (b) $y = c_1 e^x + c_2 e^{-x} + c_3 e^{2x}$
(c) $y = c_1 e^x + (c_2 + c_3 x) \sin x$ (d) $y = c_1 \sin x + c_2 \cos x$
59. $\frac{1}{D-1} e^x =$
(a) e^x (b) $e^x - x$

- (c) xe^x (d) e^{2x}
60. $D(\cos 3x) =$
 (a) $-\sin 3x$ (b) $-3 \sin 3x$
 (c) $\cos(-3x)$ (d) $-3 \cos 3x$
61. $\frac{1}{D}(\cos x) =$
 (a) $\sin x$ (b) $-\sin x$
 (c) $\cos x$ (d) $\cos^2 x$
62. If differential equation is $f(D) = a$, then y_p is of the form
 (a) $x^k (A \cos \alpha x + B \sin \alpha x)$ (b) Ax^k
 (c) $Ax^k e^x$ (d) none
63. If differential equation is of the form $f(D) = a \cos \alpha x$ then y_p is of form
 (a) Ax^k (b) $Ax^k e^x$
 (c) $x^k (A \cos \alpha x + B \sin \alpha x)$ (d) $Ax^k \cos \alpha x$
64. The equation of the form $a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = F(x)$ is:
 (a) Ricatte's equation (b) Lagrange's equation
 (c) Cauchy-Euler equation (d) Bernoulli equation
65. If y_1, y_2 are two differentiable functions of x on $[a, b]$ then $\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is called:
 (a) distance (b) determinant
 (c) wronskian (d) multiplier

KEY to MCQ of Differential Equations

1	a	2	a	3	b	4	b	5	b	6	c
7	c	8	a	9	b	10	c	11	b	12	b
13	c	14	d	15	d	16	c	17	c	18	b
19	a	20	c	21	d	22	b	23	c	24	b
25	d	26	d	27	c	28	a	29	c	30	c
31	b	32	d	33	b	34	d	35	a	36	d
37	b	38	d	39	d	40	c	41	b	42	b
43	c	44	c	45	b	46	c	47	c	48	c
49	b	50	c	51	a	52	b	53	b	54	c
55	c	56	c	57	c	58	a	59	c	60	b
61	b	62	c	63	c	64	c	65	c		